Operator Name: KRIISH HATE   Started on: 09/10/2024

09/06/2024

Task: Project meeting with Ruifeng

Things that need to be worked on:

↳ MOSFET TESTING
- IRF246NPBF
- 2N7000

↳ Housing finalize

↳ PCB Design finalise

09/10/2024

Task: Checking power consumption of air unit with all the peripherals attached.

Peak current consumption without cellular module: 35 mA

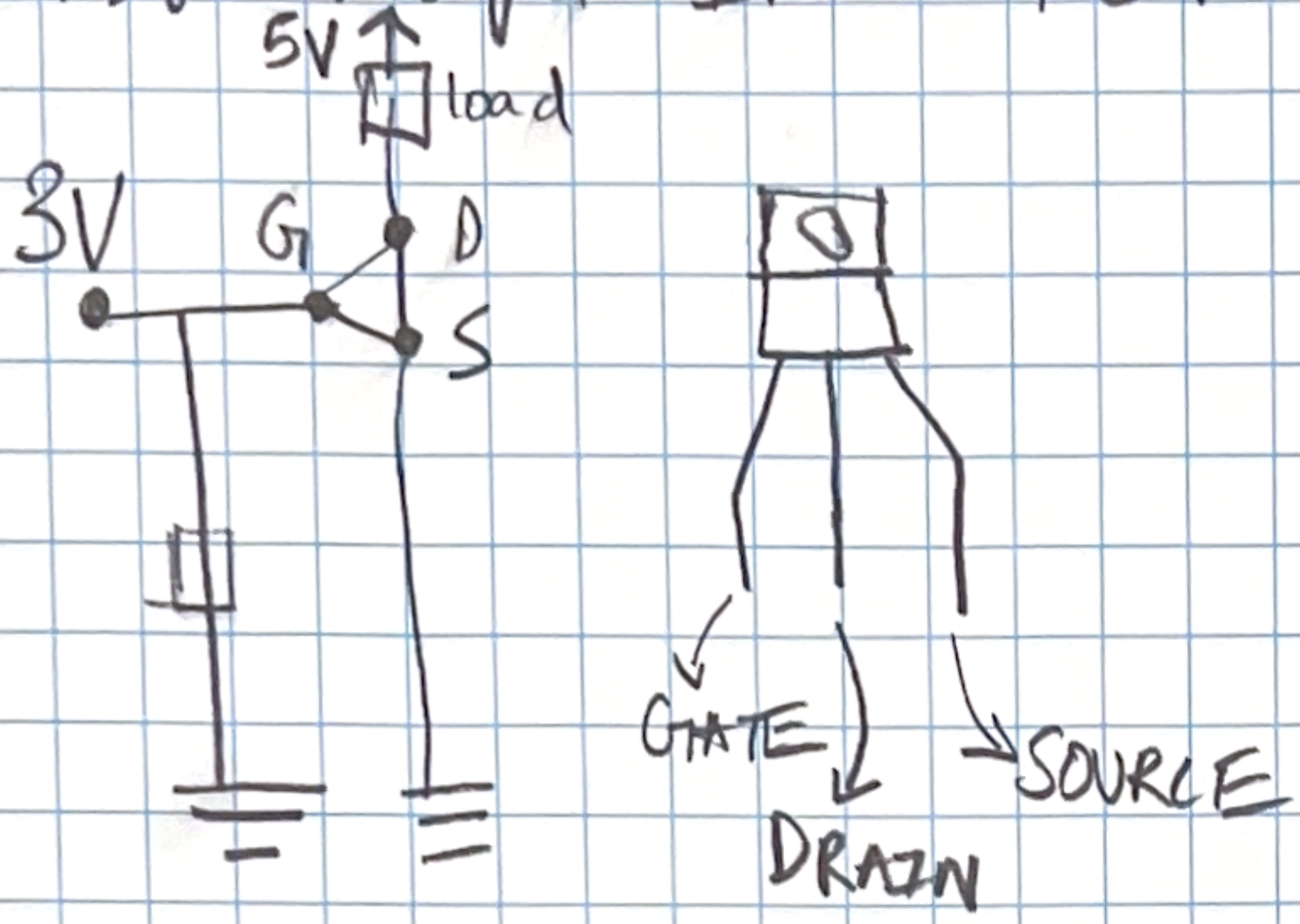
Peak current consumption with cellular module: 135 mA (When sending data)

Peak current consumption with cellular module: 60 mA (Without sending data)

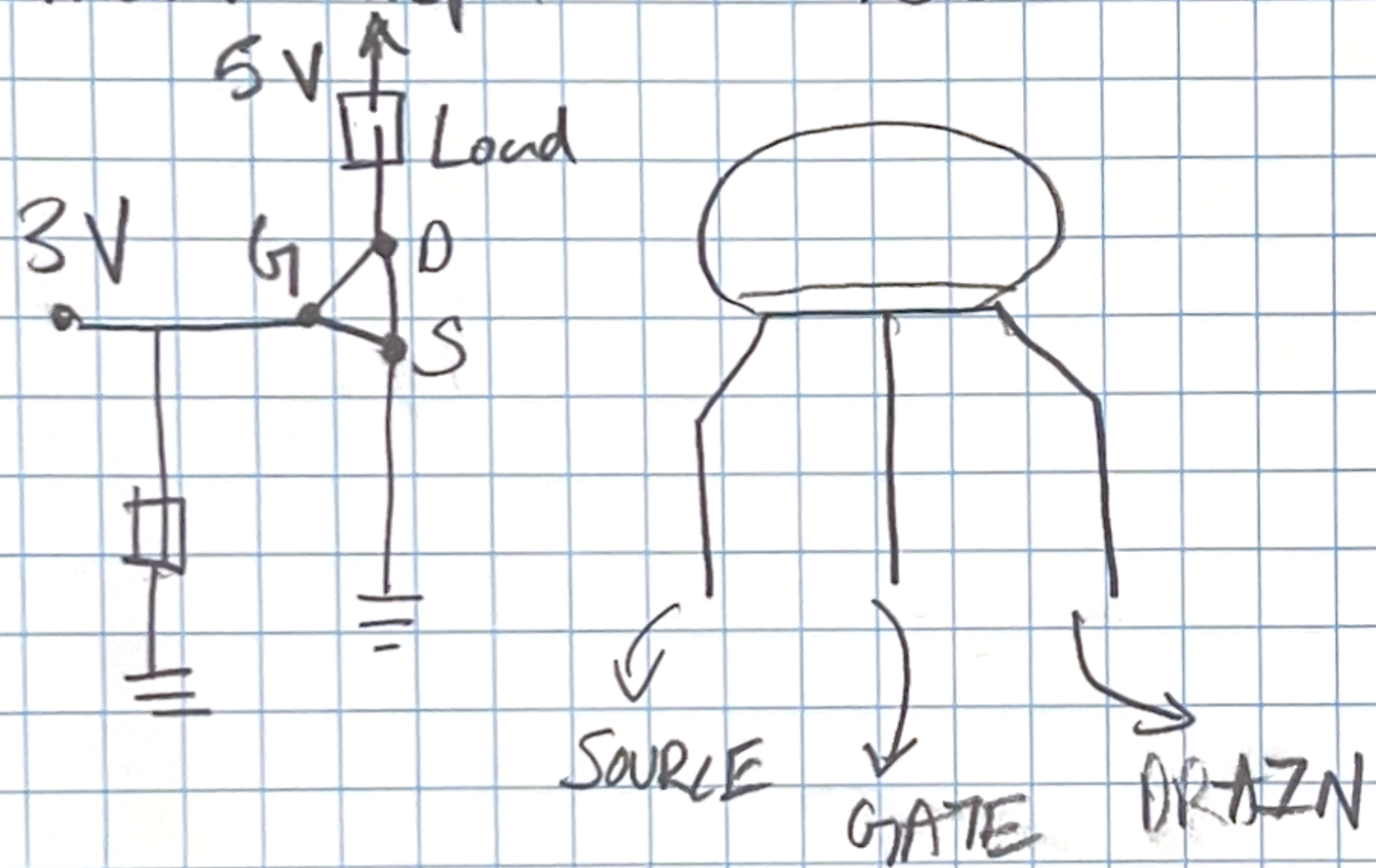
Based on peak consumption at 135 mA for the whole system.

2N7000 Rated for a maximum drain current of 200 mA (Pg2 of Datasheet)

Bigger Mosfet IRF246NPbF N-Chanel



Smaller Mosfet 2N7000 N-Chanel



09/16/2024

Task: Measuring the dimensions of the current enclosure and finalizing a suitable enclosure which is <u>waterproof</u> & <u>UV resistant</u>.

Current dimensions which fits all the components + the battery pack

L 6" inch × B 4.5" inch × h 3" inch

Potential Candidates chosen from (Refer to the Links)

↳ Amazon

↳ McMaster Carr

09/17/2024

Task: Finalising another MOSFET as the power consumption of the particle could exceed 500mA under certain conditions

→ Go for a 1 Amp rated MOSFET to be on the safer side

Finalised MOSFET NChanel 2SK3447-T12-E (Refer to link on the E Lab book)

Task: ~~Finding out~~ the peak current draw from the SD card module.

150mA (Based on the link)

Should I use the 200 mA rated MOSFET for this case?

Task: Finalize the box clamping mechanism

10/13/2024

Number of Male headers		Number of female headers	
Arduino	36	Arduino	36
LTE Module	0	LTE Module	28
RTC	0	RTC	8
SD card	0	SD card	8